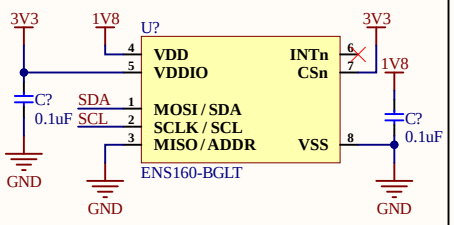
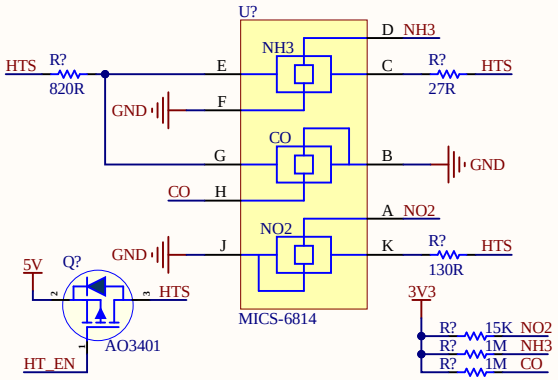
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Size A	Number	Revision
Date: 6-09-2023		Sheet of
File: D:\eSpark-Consultants\..\Top-Sheet.SchDoc		Drawn By:

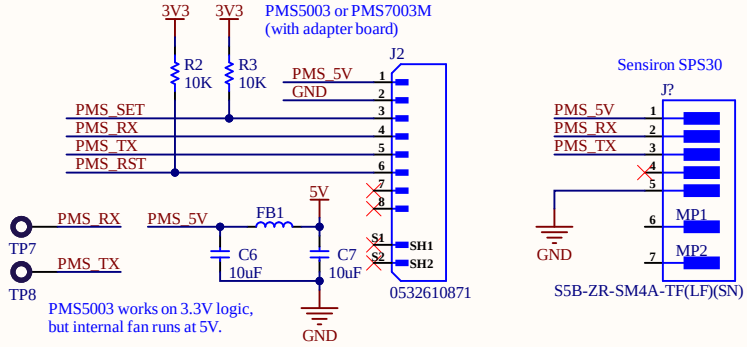
eCO2 TVOC AQI SENSOR



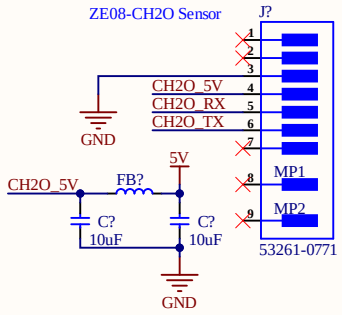
SUPPPORT SENSOR TO DETECT MARIJUANA



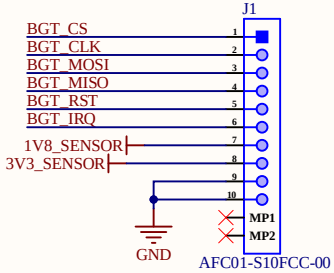
PM2.5 ENVIRONMENT SENSOR CIRCUIT



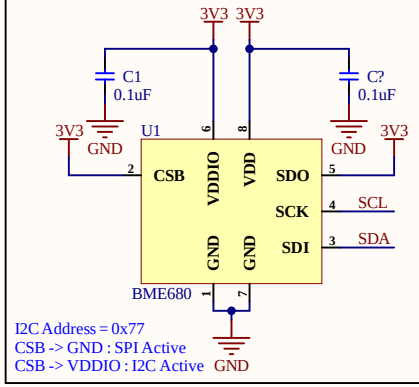
SUPPPORT SENSOR FOR MARIJUANA



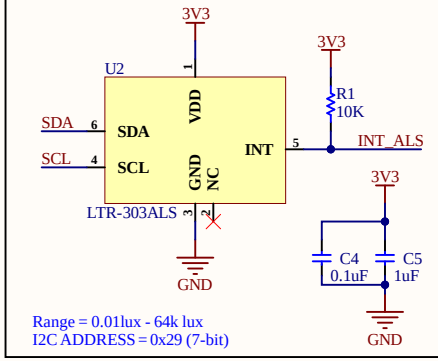
BGT60 RADAR SENSOR



TEMP HUM PRESSURE SENSOR



AMBIENT LIGHT SENSOR CIRCUIT



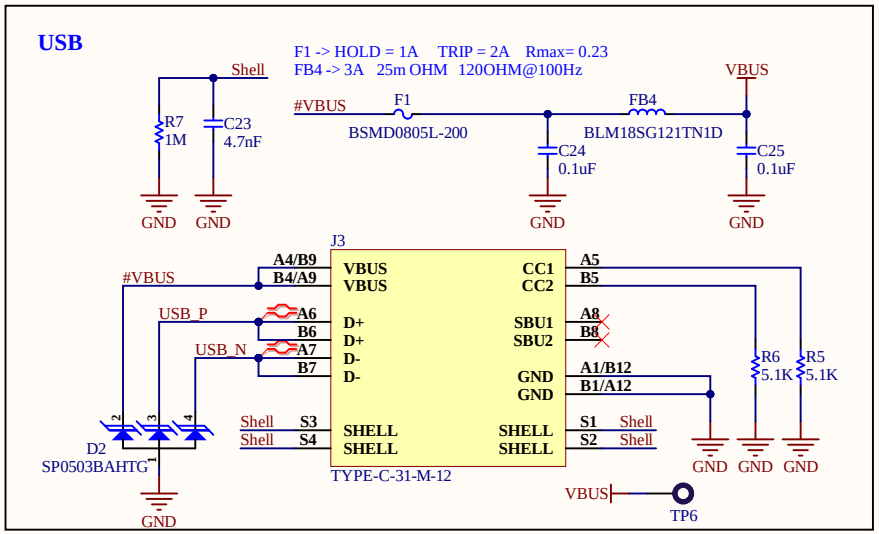


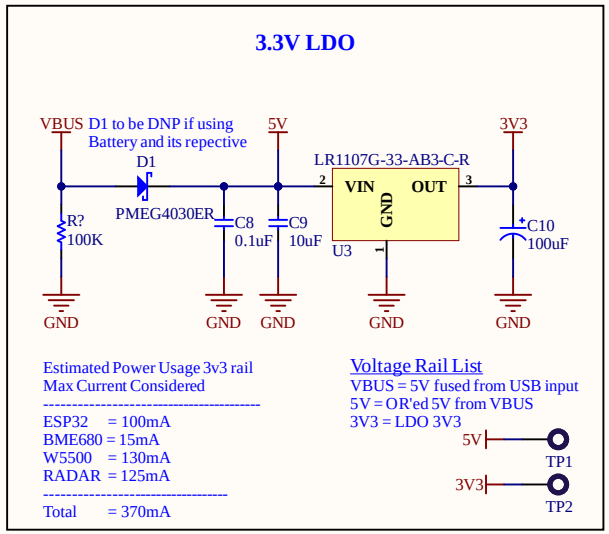
I2C pullups are given on Microcontroller-Sheet

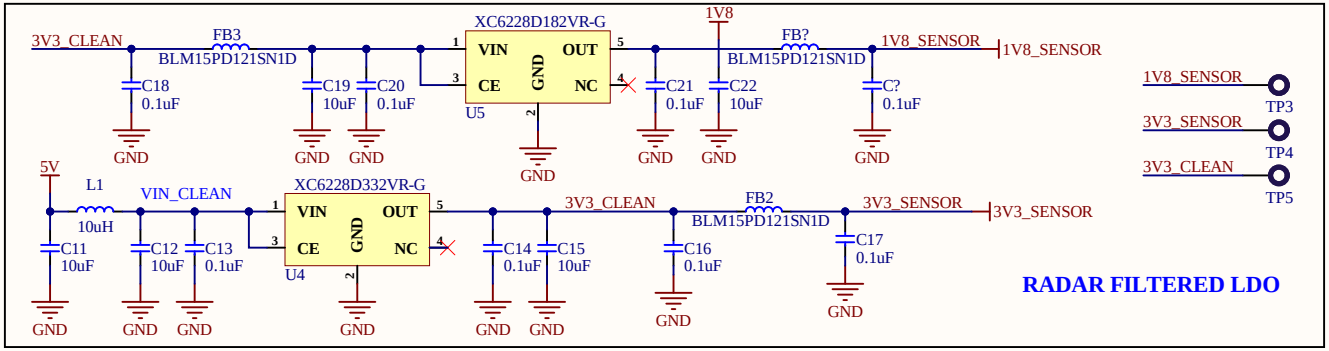
Sensors Section

eSpark Consulting Group

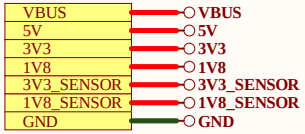
Title: Sensor-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 09-Jun-23	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 6







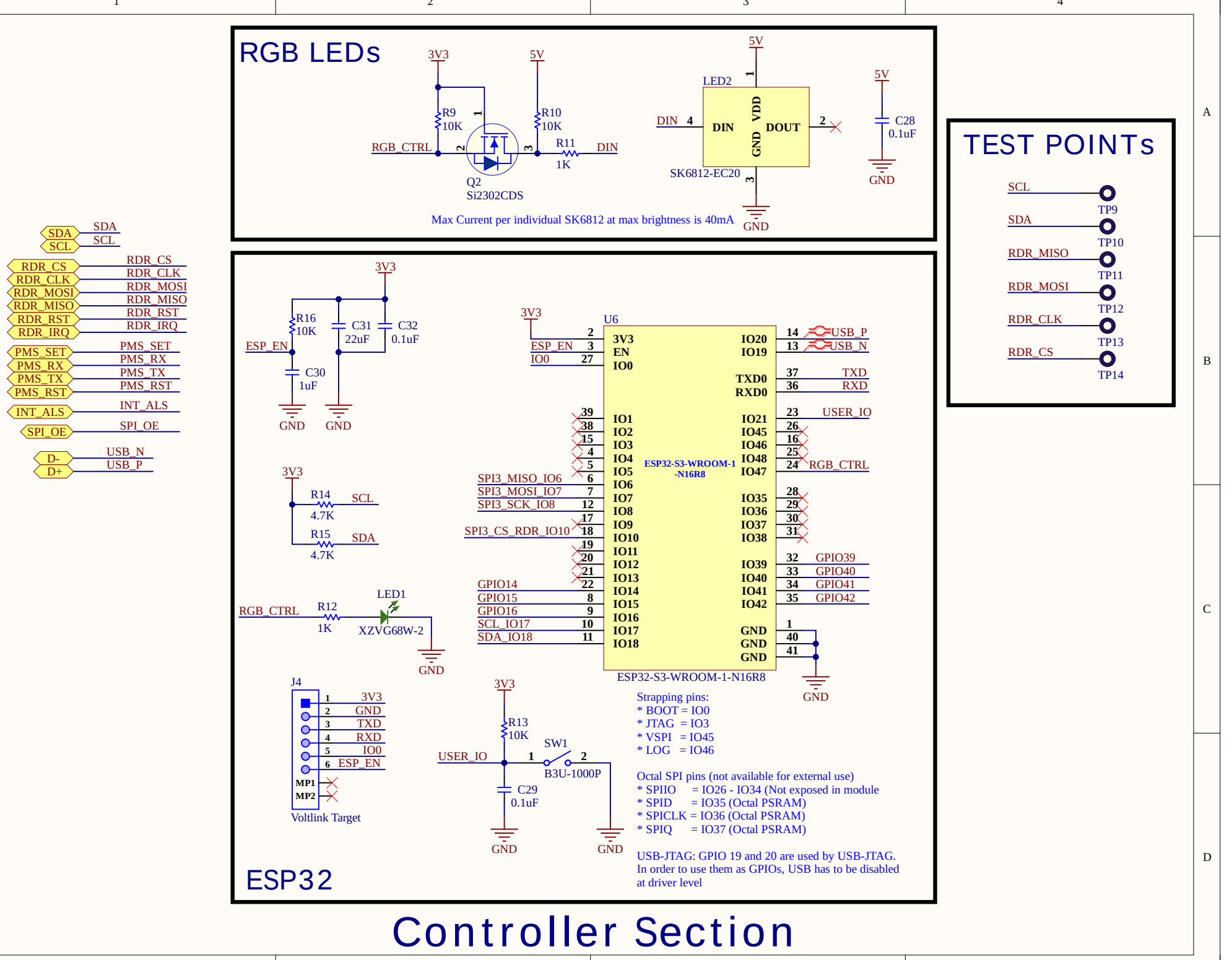


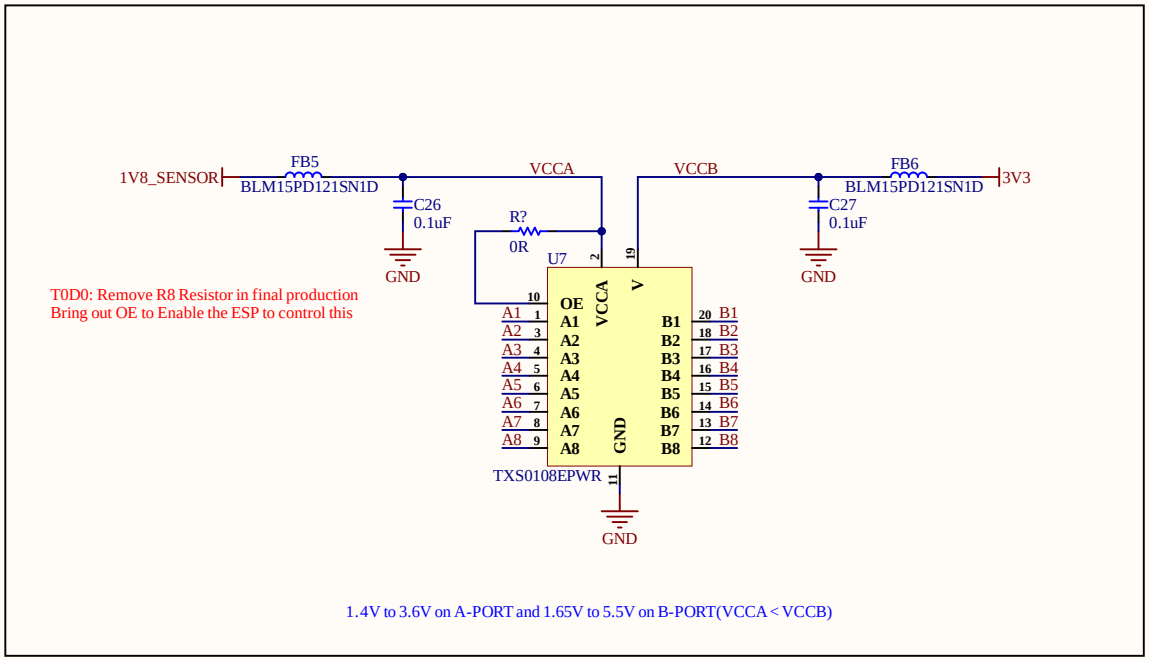


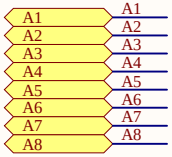
Power Section

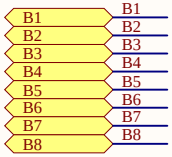
eSpark Consulting Group

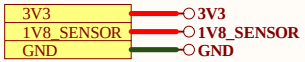
Title: Power-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 09-Jun-23	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 6


RGB LEDs
3V3
5V
R9
10K
R10
10K
RGB_CTRL
R11
1K
DIN
Q2
Si2302CDS
LED2
DIN
VDD
DOUT
GND
SK6812-EC20
C28
0.1uF
Max Current per individual SK6812 at max brightness is 40mA
TEST POINTs
SCL
TP9
SDA
TP10
RDR_MISO
TP11
RDR_MOSI
TP12
RDR_CLK
TP13
RDR_CS
TP14
SDA
SCL
RDR_CS
RDR_CLK
RDR_MOSI
RDR_MISO
RDR_RST
RDR_IRQ
PMS_SET
PMS_RX
PMS_TX
PMS_RST
INT_ALS
SPI_OE
D-
D+
USB_N
USB_P
R16
10K
C31
22uF
C32
0.1uF
C30
1uF
ESP_EN
R14
4.7K
R15
4.7K
U6
ESP32-S3-WROOM-1
-N16R8
IO0
TXD0
RXD0
USB_P
USB_N
TXD
RXD
USER_IO
RGB_CTRL
SPI3_MISO_IO6
SPI3_MOSI_IO7
SPI3_SCK_IO8
SPI3_CS_RDR_IO10
GPIO14
GPIO15
GPIO16
SCL_IO17
SDA_IO18
GPIO39
GPIO40
GPIO41
GPIO42
LED1
R12
1K
XZVG68W-2
J4
MP1
MP2
Voltlink Target
R13
10K
SW1
B3U-1000P
C29
0.1uF
Strapping pins:
* BOOT = IO0
* JTAG = IO3
* VSPI = IO45
* LOG = IO46
Octal SPI pins (not available for external use)
* SPIIO = IO26 - IO34 (Not exposed in module
* SPID = IO35 (Octal PSRAM)
* SPICLK = IO36 (Octal PSRAM)
* SPIQ = IO37 (Octal PSRAM)
USB-JTAG: GPIO 19 and 20 are used by USB-JTAG.
In order to use them as GPIOs, USB has to be disabled
at driver level
ESP32
Controller Section





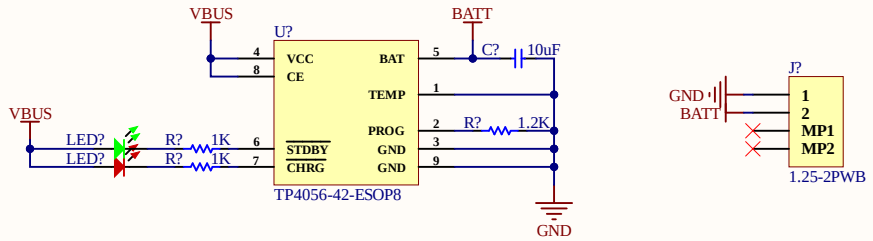




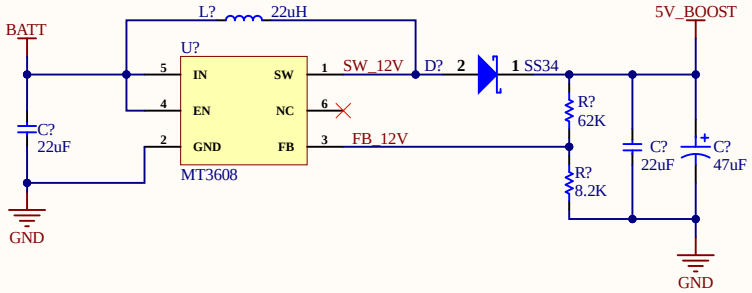
Level Shifter Section

Title: Level_Shifter_Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 09-Jun-23	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 6

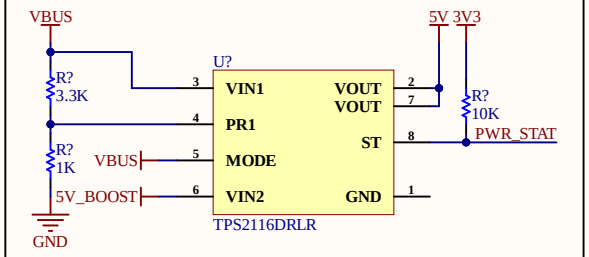
BATTERY CHARGING CIRCUIT



STEP UP CONVERTER



POWER MUX PRIORITY SWITCHOVER



Title: Charging-Boost-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 09-Jun-23	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 6

Comment	Description	Designator	Footprint	LibRef	Quantity
0.1uF	CAP CER 0.1UF 50V X5R 0402, CAP CER 0.1UF 10V X5R 0402	C1, C4, C8, C13, C14, C16, C17, C18, C20, C21, C24, C25, C26, C27, C28, C29, C32, C7	Cap-0402	C1005X5R1H104K050B B, CC0402KRX5R6BB104	21
1uF	CAP CER 1UF 10V X5R 0402	C5, C30	Cap-0402	CC0402KRX5R6BB105	2
10uF	CAP CER 10UF 10V X5R 0402	C6, C7, C9, C?	Cap-0402	CC0402MRX5R6BB106	6
100uF	CAP TANT 100UF 20% 10V 1411	C10	Case-B	TAJB107M010RNJ	1
10uF	CAP CER 10UF 10V X5R 0603	C11, C12, C15, C19, C22	Cap-0603	CC0603KRX5R6BB106	5
4.7nF	CAP CER 4700PF 1KV 17R 1206	C23	Cap-1206	CC1206KKX7RCBB472	1
22uF	CAP CER 22UF 10V X5R 0603	C31	Cap-0603	CC0603MRX5R6BB226	1
22uF	25V 22uF X5R ±20% 0805 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C7	Cap-0805	CS2012X5R226M25NRE	2
47uF	CAP ALUM POLY 47UF 20% 25V SMD	C9	25SVPF47M	25SVPF47M	1
PMEG4030ER	SOD-123FL Schottky Barrier Diodes (SBD) ROHS	D1	SOD-123FL	PMEG4030ER	1
SP0503BAHTG	ESD Suppressor TVS 5.5V 4-Pin(3+Tab) SOT-143 T/R	D2	SOT-143-4_	SP0503BAHTG	1
SS34	0.5mA@40V 40V Single 3A 0.55V@3A SMA(DO-214AC) Schottky Barrier Diodes (SBD) ROHS	D7	SMA	SS34	1
BSMD080SL-200	8V 2A 50A 1W -40?C~+85?C 4A 4mΩ SL 45mΩ 0805 Resettable Fuses ROHS	F1	Fuse-0805	BSMD080SL-200	1
BLM18G121TN1D	FERRITE BEAD 120 OHM 0603 1LN	FB1, FB4, FB?	IND-0603 (1608 Metric)	BLM18G121TN1D	3
BLM15PD1213SN1D	FERRITE BEAD 120 OHM 0402 1LN	FB2, FB3, FB5, FB6, FB?	IND-0402	BLM15PD1213SN1D	5
AFC01-S10FCC-00	Clamshell 10 Bottom Contact Surface Mount 0.5mm SMD,P=0.5mm FFC/FPC Connectors ROHS	J1	AFC01-S10FCC-00	AFC01-S10FCC-00	1
0532610871	CONN HEADER SMD R/A 8POS 1.25MM	J2	53261-0871	0532610871	1
TYPE-C-31-M-12	250mA 1 16 Female Type-C SMD USB Connectors ROHS	J3	TYPE-C-31-M-12_	TYPE-C-31-M-12	1
Voltlink Target	1x6P SH 1.1mm 6 SMD,P=1mm Wire To Board / Wire To Wire Connector ROHS	J4	Voltlink Target	C145963	1
1.25-2PWB	1x2P 1 1.25mm Male pin 2 -40?C ~+85?C Surface Mount SMD,P=1.25mm Wire To Board / Wire To Wire Connector ROHS	J7	1.25-2PWB	1.25-2PWB	1
53261-0771	CONN HEADER SMD R/A 7POS 1.25MM	J7	53261-0771	53261-0771	1
SSB-2R-SM4A-TF(LF)(SN)	CONN HEADER SMD R/A 5POS 1.5MM	J7	SSB-2R-SM4A-TF(LF)(SN)	SSB-2R-SM4A-TF(LF)(SN)	1
10uH	FIXED IND 10UH 1A 540 MOHM SMD	L1	IND-1008 (2520 Metric)	VLS252012HBX-100M-1	1
22uH	22uH ±20% SMD Power Inductors ROHS	L7	SMRH74-220MT	SMRH74-220MT	1
XZVG68W-2	LED GREEN CLEAR CHIP SMD	LED1	LED-0402	XZVG68W-2	1
SK6812-EC20	5mA ~40?C~+80?C RGB SMD,2x2mm Light Emitting Diodes (LED) ROHS	LED2	SK6812-EC20	SK6812-EC20	1
Emerald	Emerald 0603 Light Emitting Diodes (LED) ROHS	LED?	LED-0603	NCD0603G2	1
RED	RED LED 0603 Light Emitting Diodes	LED?	LED-0603	NCD0603R3	1
SI2302CDS	N-CHANNEL 20-V (D-S) MOSFET	Q2	Sot-23	SI2302CDS	1
AO3401	Trans Mosfet P-ch 30V 4A 3-PIN SOT-23 T/r	Q7	SOT-23	AO3401	1
10K	RES 10K OHM 1% 1/10W 0402	R1, R2, R3, R9, R10, R16, R?	Res-0402	RC0402FR-0710KL	7
5.1K	RES 5.1K OHM 1% 1/16W 0402	R5, R6	Res-0402	RC0402FR-075K1L	2
1M	RES SMD 1M OHM 1% 1/10W 0603	R7, R?	Res-0603	RC0603FR-071ML	3
1K	RES SMD 1K OHM 1% 1/10W 0603	R11, R12	Res-0603	RC0603FR-071KL	2
10K	RES SMD 10K Ohm 1% 1/10W 0603	R13	Res-0603	RC0603FR-0710KL	1
4.7K	RES SMD 4.7K OHM 1% 1/10W 0603	R14, R15	Res-0603	RC0603FR-074K7L	2
0R	RES SMD 0R, 1%, 0.1W, 0603	R?	Res-0603	RC0603FR-070RL	1
1K	RES 1K OHM 1% 1/16W 0402	R?	Res-0402	RC0402FR-071KL	3
1.2K	RES 1.2K OHM 1% 1/16W 0402	R?	Res-0402	RC0402FR-071K2L	1
3.3K	RES 3.3K OHM 1% 1/16W 0402	R?	Res-0402	RC0402FR-073K3L	1
8.2K	RES SMD 8.2K OHM 0.1% 1/10W 0603	R?	Res-0603	RT0603BRD078K2L	1
15K	RES 15K OHM 1% 1/10W 0603	R?	Res-0603	RC0603FR-0715KL	1
27R	RES 27 OHM 1% 1/10W 0603	R?	Res-0603	RC0603FR-0727RL	1
62K	RES SMD 62K OHM 0.1% 1/10W 0603	R?	Res-0603	RT0603BRD0762KL	1
100K	RES 100K OHM 1% 1/16W 0402	R?	Res-0402	RC0402FR-07100KL	1
130R	RES 130 OHM 1% 1/10W 0603	R?	Res-0603	RC0603FR-07130RL	1
820R	RES 820 OHM 1% 1/10W 0603	R?	Res-0603	RC0603FR-07820RL	1
B3U-1000P	SWITCH TACTILE SPST-NO 0.05A 12V	SW1	B3U-1000P	B3U-1000P	1
TEST POINT	Test Point ON	TP1, TP2, TP3, TP4, TP5, TP6, TP7, TP8, TP9, TP10, TP11, TP12, TP13, TP14	TEST POINT	TEST POINT	14
BME680	SENSOR RH PRESSURE TEMP VOC	U1	BME680	BME680	1
LTR-303ALS	SENSOR OPT AMBIENT 6DFN	U2	6-Pin ChipLED	LTR-303ALS-01	1
LR1107G-33-AB3-C-R	600mA 60dB Fixed 3.3V~3.3V Positive 7V SOT-89-3 Linear Voltage Regulators (LDO) ROHS	U3	SOT-89-3	LR1107G-33-AB3-C-R	1
XC6228D332VR-G	300mA 80dB@(1kHz) 305mV@(300mA) Fixed 3.3V~3.3V Positive 1.5.5V SOT-25-5 Linear Voltage Regulators (LDO) ROHS	U4	SOT-25-5	XC6228D332VR-G	1
XC6228D182VR-G	300mA 80dB@(1kHz) 410mV@(300mA) Fixed 1.8V~1.8V Positive 1.5.5V SOT-25-5 Linear Voltage Regulators (LDO) ROHS	U5	SOT-25-5	XC6228D182VR-G	1
ESP32-S3-WROOM-1-N16R8	SMD MODULE, ESP32-S3R8 WiTh 3.3V	U6	41-SMD Module	ESP32-S3-WROOM-1-N16R8	1
TXS0108EPWR	IC TRANSLATOR BIDIR 20TSSOP	U7	TSSOP-20	TXS0108EPWR	1
ENS160-BGLT	Digital Metal-Oxide Multi-Gas Sensor	U7	ENS160-BGLT	ENS160-BGLT	1
MICS-6814	CO/NO2/NH3 Triple Gas Sensor Metaloxide Gas Detection Sensor SMD T/R	U7	MICS-6814	MICS-6814	1
MT3608	boost converterType 1 Boost 28V 2V~24V 1.2MHz SOT-23-6 DC-DC Convertors ROHS	U7	SOT-23-6	MT3608	1
TP4056-42-ESOP8	ESOP-8 Battery Management ICs ROHS	U7	ESOP-8	TP4056-42-ESOP8	1
TPS2116DRLR	1.6-V TO 5.5-V, 40-M, 2.5-A, LOW	U7	SOT-583	TPS2116DRLR	1